

PROJECT SPECIFICATIONS

Project Title:

AI – Powered Employee Attrition Prediction and Workforce Intelligence System

Group Members:

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1. User and System Specifications

Component	Specification
Users	HR Manager, HR Executive, Administrator
Operating System	Windows 10/11 or Linux
Frontend	Streamlit
Backend	Flask/FastAPI
Database	SQLite
Programming Language	Python 3.x

2. Input Specifications

Input	Description
Employee Details	Age, Department, Job Role
Salary Information	Monthly Income
Work Information	Overtime, Years at Company
Performance Details	Performance Rating
Satisfaction Metrics	Job Satisfaction, Work-Life Balance
Experience	Total Working Years

3. Output Specifications

Output	Description
Attrition Prediction	Stay or Leave
Risk Score	Probability of employee attrition
Prediction Explanation	Key factors influencing prediction
Retention Recommendations	Suggested actions for HR
Manager Decision Summary	AI-generated summary
HR Dashboard	Workforce analytics and visual reports

4. Model Specifications

Component	Specification
Prediction Model	Artificial Neural Network (ANN)
Alternative Models	Random Forest, XGBoost
Explainable AI	SHAP
Target Variable	Attrition (Yes/No)
Dataset	IBM HR Employee Attrition Dataset
Evaluation Metrics	Accuracy, Precision, Recall, F1-Score

5. Dataset Specifications:

Parameter	Specification
Dataset Name	IBM HR Employee Attrition Dataset
Dataset Type	Structured CSV
Number of Records	1470
Number of Features	35
Target Attribute	Attrition (Yes/No)

6. Functional Module Specifications

Module	Function
Employee Data Management	Accepts and validates employee information.
Attrition Prediction	Predicts whether an employee is likely to leave.
Explainable AI	Displays the key reasons behind each prediction.
Smart Recommendations	Suggests retention strategies for HR.
Manager Decision Summary	Generates an AI-based summary for managers.
HR Dashboard	Visualizes attrition trends and workforce analytics.